

UNIVERSITY OF TORONTO MISSISSAUGA

CSC311 Winter 2025 Test #1 CSC311H5

Introduction to Machine Learning

Duration: 90 minutes

Instructor(s): Anthony Bonner, Bilal Taha, Marina Tawfik, Mai Ha Vu, Lisa Zhang

Test Aids: One page singled sided aid sheet. Non-programmable calculator.

Please read the following guidelines carefully.

- Please **print** your name and student number above.
- **DO NOT** write anything in the QR code area.
- **DO NOT** open or turn over the test paper until the test has started.
- This test has **6** questions. There are a total of **12 pages, DOUBLE-SIDED**.
- Answer questions clearly and completely. Provide justification **unless explicitly asked not to**.

Question	Out of
Q1	3
Q2	6
Q3	7
Q4	5
Q5	6
Q6	5
Total	32

Use this page for rough work. If you want work on this page to be marked, please indicate this clearly *at the location of the original question*.

1. **[3 marks]** For each question, circle a single best answer. No justification is required.

- | | |
|---|--|
| (a) Which of the following is true about the various types of machine learning problems? | <p>(A) Reinforcement learning (RL) problems tend to be easier to solve than supervised learning problems, since there are many possible “good” actions in RL problems and fewer ethical issues to consider.</p> <p>(B) Typical examples of unsupervised learning include game-playing: i.e., deciding what action to take in each time step in a game like Chess, Go, Atari, or Tetris.</p> <p>(C) Supervised learning problems include classification, regression, and clustering.</p> <p>(D) All of the above.</p> <p>(E) None of the above.</p> |
| (b) Which of the following is true about the effect of normalizing the training data on model accuracy? | <p>(A) Normalization has no effect on the accuracy of a k-NN model.</p> <p>(B) Normalization generally improves a linear model due to the curse of dimensionality.</p> <p>(C) Normalization generally improves a decision tree model by scaling gradient updates appropriately.</p> <p>(D) All of the above.</p> <p>(E) None of the above.</p> |
| (c) Which of the following best describes the purpose of the test set? | <p>(A) To decide which model family is the best (e.g., k-nearest neighbour vs decision trees).</p> <p>(B) To tune model settings that cannot be easily tuned using an optimizer during training.</p> <p>(C) To test whether the curse of dimensionality holds.</p> <p>(D) To see how well a model performs on new data not in the training and validation sets.</p> <p>(E) Both (A) and (D).</p> |

2. [6 marks] For each of the following quantities, circle whether **increasing** the quantity would make the model **more likely to overfit**, **more likely to underfit**, or **neither**. No explanation is required.

(a) The maximum depth of a decision tree classifier.

Circle one: More likely to overfit or More likely to underfit or Neither

(b) The weight of the regularizer λ in the regularized cost function $\mathcal{E}_{\text{reg}}(\mathbf{w}) = \mathcal{E}(\mathbf{w}) + \frac{\lambda}{2} \sum_{j=0}^D w_j^2$ of a linear model, where $\mathbf{w}$ is the weight vector and $\mathcal{E}$ is the unregularized cost function (e.g., the mean-squared error).

Circle one: More likely to overfit or More likely to underfit or Neither

(c) The value of k in a k-Nearest Neighbour model.

Circle one: More likely to overfit or More likely to underfit or Neither

(d) The number of data points in our training set.

Circle one: More likely to overfit or More likely to underfit or Neither

(e) The number of new features added when using polynomial feature mapping, e.g., in a linear model.

Circle one: More likely to overfit or More likely to underfit or Neither

(f) The number of data points in the test set.

Circle one: More likely to overfit or More likely to underfit or Neither

3. **[7 marks]** Suppose that we would like to train a linear regression model $y = \mathbf{w}^T \mathbf{x}$ using gradient descent, where

$\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_D \end{bmatrix}$ is the weight vector and $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_D \end{bmatrix}$ is the vector of input features. We would like to use the following

loss function $\mathcal{L}_1(y, t) = |y - t|$ to train this model.

- (a) **[3 marks]** Compute the derivative $\frac{\partial \mathcal{L}_1}{\partial w_j}$. Show all your work.

- (b) **[2 marks]** Vectorize your results above to write down $\nabla_{\mathbf{w}} \mathcal{L}_1$ using only vector quantities. Please define any new mathematical notation that you introduce.

- (c) [**2 marks**] Explain how the gradient $\nabla_{\mathbf{w}} \mathcal{L}_1$ that you calculated would be used to learn an optional set of weights $\mathbf{w}$ for our model (hypothesis). Answer this question by outlining the learning algorithm and write down the update rule—as we are looking for an understanding of how linear models like these are trained. Please either use the standard notation that we used in class, or define any new mathematical notation that you introduce.

4. [5 marks] Suppose that we encounter the following joint distribution, where X and Y are binary random variables:

$$P(X = 0, Y = 0) = 0.2$$

$$P(X = 0, Y = 1) = 0.2$$

$$P(X = 1, Y = 0) = 0.2$$

$$P(X = 1, Y = 1) = 0.4$$

- (a) [1 mark] Compute $H(X)$, the entropy of X . Please include 3 decimal places.
- (b) [1 mark] Compute $H(X|Y = 0)$, the conditional entropy of X given $Y = 0$. Please include 3 decimal places.
- (c) [1 mark] Compute $H(X|Y = 1)$, the conditional entropy of X given $Y = 1$. Please include 3 decimal places.
- (d) [1 mark] Compute $H(X|Y)$, the expected conditional entropy of X given Y . Please include 3 decimal places.
- (e) [1 mark] Recall that $IG(X|Y) = H(X) - H(X|Y)$ is the information gain of X when observing Y . Explain, in plain English, what this quantity is intended to measure.

5. **[6 marks]** During the lecture, we explored a few approaches for modeling a classification problem. Clearly explain why each of the below models/approaches is problematic, and thus cannot be trained using gradient descent. In each case we use $\mathbf{x}$ to represent the vector of input features, $\mathbf{w}$ to represent the vector of weights, t to represent the ground-truth label (with $t \in \{0, 1\}$), and $\mathbb{I}[e]$ to represent the indicator of e (i.e., $\mathbb{I}[e] = 1$ if e is true, and $\mathbb{I}[e] = 0$ otherwise).

(a) **[2 marks]** Using linear regression and square loss, i.e.

$$\begin{aligned} z &= \mathbf{w}^T \mathbf{x} \\ y &= \mathbb{I}[z \geq 0] \\ \mathcal{L}_{SE} &= (z - t)^2 \end{aligned}$$

(b) **[2 marks]** Using a threshold activation function and 0-1 loss, i.e.

$$\begin{aligned} z &= \mathbf{w}^T \mathbf{x} \\ y &= \mathbb{I}[z \geq 0] \\ \mathcal{L}_{0-1} &= \mathbb{I}[y \neq t] \end{aligned}$$

(c) [**2 marks**] Using sigmoid activation function and square loss, i.e.

$$z = \mathbf{w}^T \mathbf{x}$$

$$y = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\mathcal{L}_{SE} = \frac{1}{2}(y - t)^2$$

6. **[5 marks]** For this question, we would like to perform classification on the following training data:

$$\mathcal{D} = \{(x^{(1)} = 1, t^{(1)} = 0), (x^{(2)} = 2, t^{(2)} = 1)\}.$$

Euclidean distance will be used in all parts of this question.

- (a) **[2 marks]** Draw the decision boundary of a 1NN model trained on this data set. Label the predictions in each region.

- (b) **[3 marks]** Suppose that we use a feature mapping function $\Psi(x) = \begin{bmatrix} x \\ x^2 \end{bmatrix}$ to first map our training data $\mathcal{D}$ into a higher-dimensional space, and then perform 1NN on these new features. What is the new decision boundary, expressed in terms of the original data space $\mathbb{R}$?

You might find it helpful to recall that the solution to a quadratic equation of the form $ax^2 + bx + c = 0$ where $a \neq 0$ is $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

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